ASE420 · INDIVIDUAL PROJECT

How Step Quest Works

From an iPhone step count to coins on a leaderboard

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THE BIG PICTURE

Three parts, one flow

1 · IPHONE SHORTCUT

Sends your steps

Reads today's steps from Apple Health and sends the number to the server.

2 · PYTHON SERVER

Does the work

Checks your token, saves the steps, and works out the coins. It runs on port 8000.

3 · FLUTTER APP

Shows the result

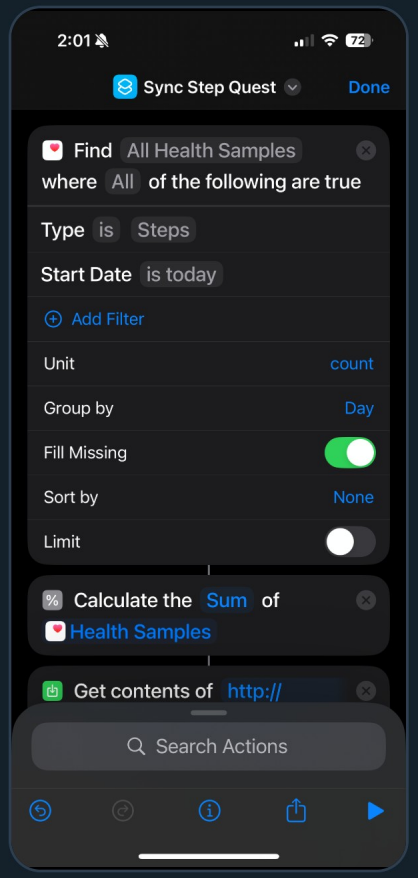
Today, Leaderboard, Friends, and Profile screens. It asks the server for everything it shows.

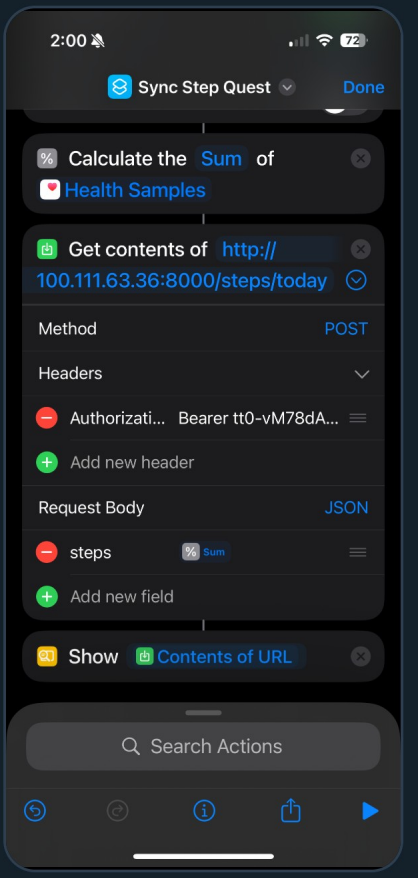
The server's /docs page lists every route and lets you try each one.

STEP 1 · THE SHORTCUT

The Shortcut reads steps and sends them

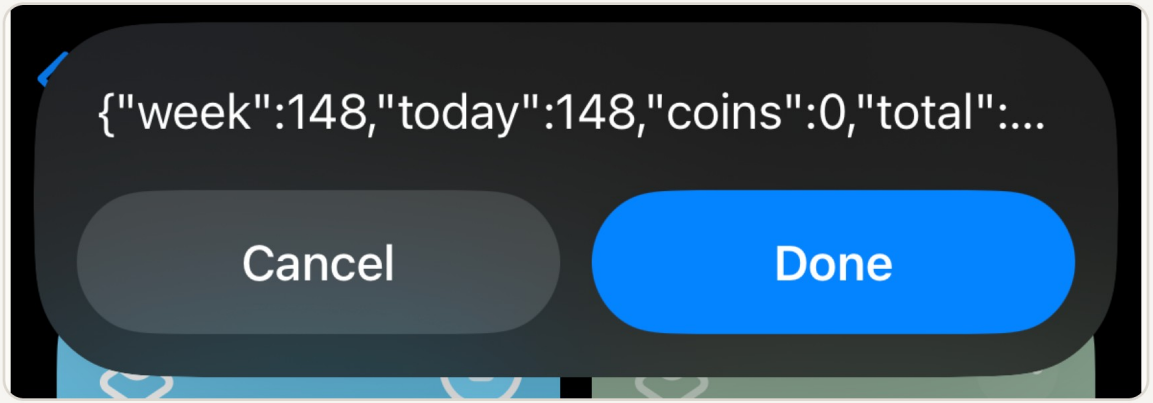
1. Find Health Samples: Type is Steps, Start Date is today, Group by Day.
2. Calculate the Sum of those samples. That is today's step count.
3. Get contents of the server URL, which ends in /steps/today. The method is POST.
4. The request carries the token in an Authorization header, and the body is {"steps": Sum}.
5. Show the server's reply.





STEP 2 · THE REPLY

The server answers with your totals



`today`
Steps for today

`week`
Steps in the last 7 days

`total`
All steps ever sent

`coins`
Total divided by 1,000, rounded down, times 10

THE /docs PAGE

Every route the server has

METHOD	ROUTE	WHAT IT DOES	LOGIN
POST	/auth/signup	Create an account and get a token	No
POST	/auth/login	Log in and get a token	No
GET	/me	Your username, steps, and coins	Yes
POST	/steps	Save steps for up to 31 days	Yes
POST	/steps/today	Save today's steps (the Shortcut uses this)	Yes
GET	/friends	Your friends and requests sent to you	Yes
POST	/friends/request	Ask someone to be your friend	Yes
POST	/friends/{id}/accept	Accept a friend request	Yes
GET	/leaderboard	You and your friends, most steps first	Yes

Login = needs the token. The padlock icons on /docs mark these routes.

ACCOUNTS

Signup, login, and the token

`POST /auth/signup`

Create an account

Send {username, password}. The username is 3 to 20 letters, numbers, or _. The password is at least 8 characters. The reply is {token}.

`POST /auth/login`

Log in

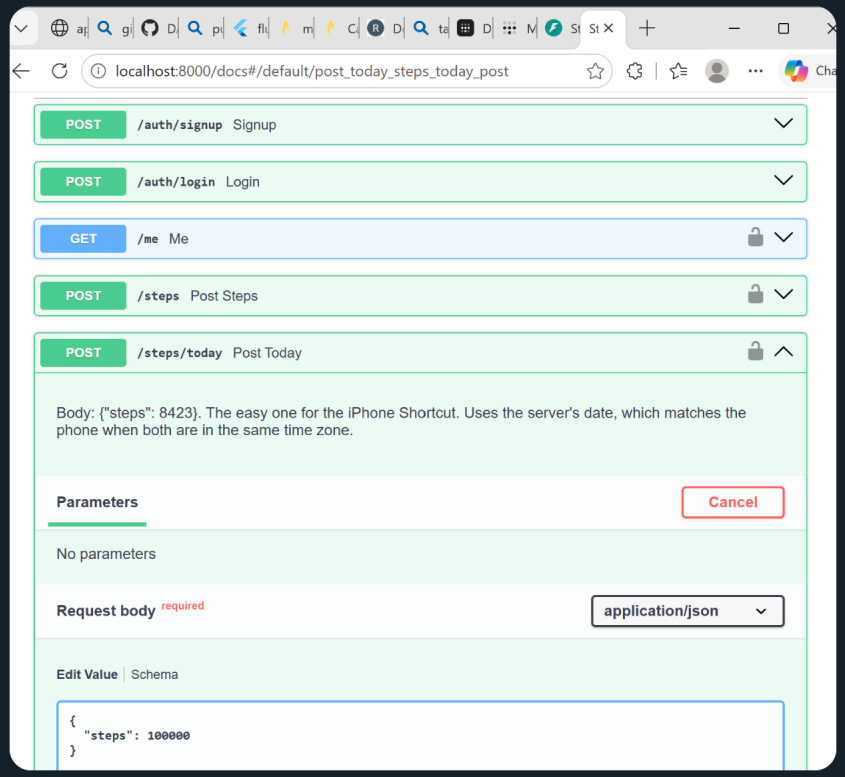
Send {username, password}. Five wrong passwords lock the username for 15 minutes. The reply is {token}.

The token proves who you are. Every other route needs it.

In /docs, click Authorize and paste only the token. The page adds Bearer for you.

SENDING STEPS

POST /steps/today and POST /steps



POST /steps/today

Body: {"steps": 8423}. One number for today. The server uses its own date, and sending again replaces the earlier number.

POST /steps

Body: {"days": [{"date": "2026-10-10", "steps": 8423}]}. Up to 31 days at once. Dates in the future are rejected.

Both accept 0 to 100,000 steps per day. The 100000 in the screenshot is that limit.

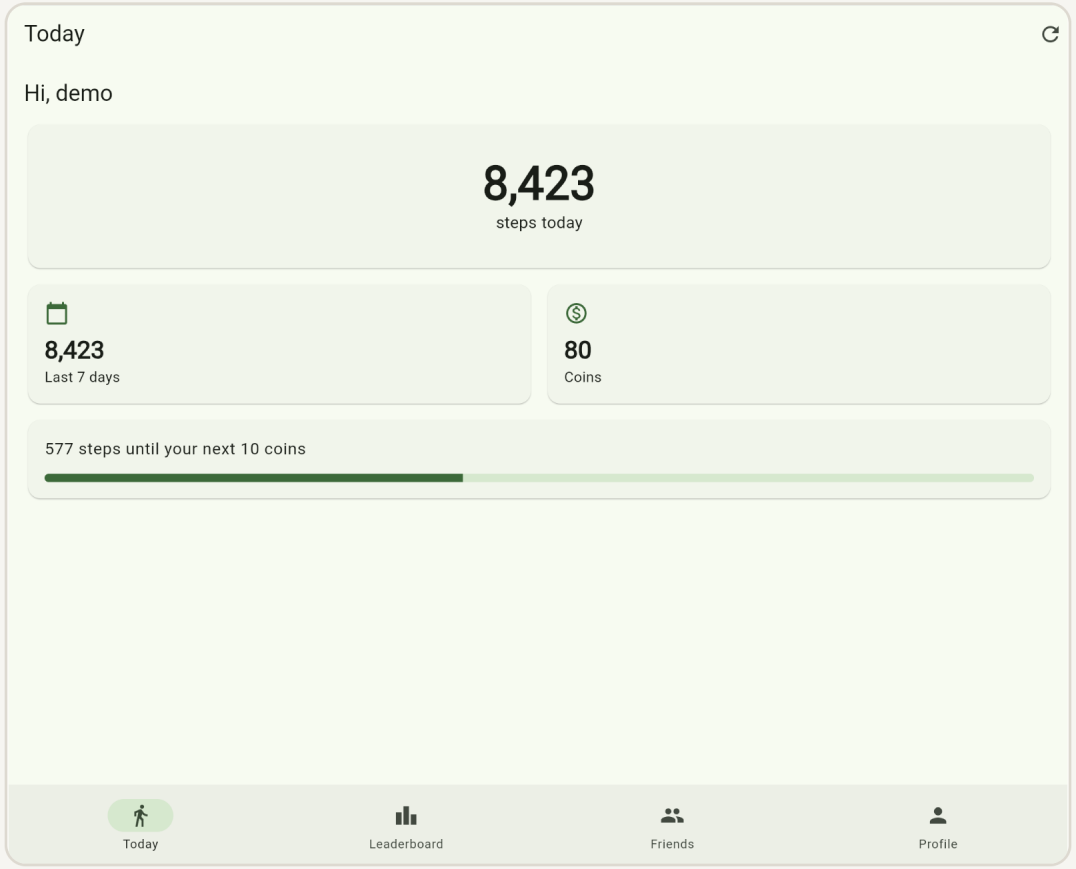
COINS

Steps become coins

coins = total steps ÷ 1,000, rounded down, × 10

Example: 8,423 steps is 8 full thousands, so 80 coins. 577 more steps earn the next 10.

The Today tab calls GET /me, which returns {username, today, week, total, coins}.



FRIENDS

Request, then accept

`POST /friends/request`

Ask

Send {username}. This creates a pending request.

`GET /friends`

Check

Returns {friends, requests}: your accepted friends, and requests people sent to you.

`POST /friends/{id}/accept`

Accept

Accepts the request from that user id. Now you are friends both ways.

If they already asked you, sending a request to them counts as accepting.

LEADERBOARD

GET /leaderboard ranks you and your friends

`?range=today`

Today

Steps from today only.

`?range=week`

Last 7 days

Steps from today and the six days before.

WHO SHOWS UP

You and friends

Accepted friends only. Most steps first, and ties go in alphabetical order. The app refreshes the list every 5 seconds.

PROFILE

The token page

```
Authorization: Bearer tt0-vM78dA…
```

- The server creates a random token when you sign up.
- The Profile tab shows it, so you can paste it into the Shortcut.
- The Shortcut sends it in the Authorization header on every request.
- Anyone with the token can send steps as you, so keep it private.
- Log out clears the login saved in the app.

TECH STACK

What each tool is for

TOOL	WHAT IT IS FOR
Python + uv	The language, and the tool that installs packages and runs the tests
FastAPI + uvicorn	Defines the routes, serves them, and builds the /docs page
SQLite	Stores users, steps, and friends in one file
Pydantic + scrypt	Checks request bodies, and hashes passwords
Flutter (Dart)	The app screens: Today, Leaderboard, Friends, Profile
iOS Shortcuts + Apple Health	Reads your steps and sends them to the server
Tailscale	A private network between the iPhone and the laptop
pytest + flutter_test	Unit, integration, regression, and acceptance tests
Marp + GitHub	Slides for the docs, and the code repository

STEP QUEST

Thank you

Any questions?

github.com/DANTHAND005/step-quest